

# Hadoop BDH Connector

*Salesforce data at 150M+ scale via the BDH data mart — filter-first, with a 24-hour lag.*

## What it is

Reads a nightly mirror of Salesforce held on the BDH Hadoop data mart (cheaper than querying Salesforce live). Because it can see 150M+ rows, its defining feature is a mandatory **filter-first** design with a fail-closed guard, so it only ever scans the data you have deliberately selected.

| Property      | Detail                                                                                                          |
|---------------|-----------------------------------------------------------------------------------------------------------------|
| Source        | BDH Hadoop data mart — nightly Salesforce mirror, ~24h behind                                                   |
| Connects with | WebHDFS (or a mounted local path); read-only HDFS principal                                                     |
| ACL model     | owner / group / public — coarser than live Salesforce sharing (a documented trade-off)                          |
| Identity      | Item id = Salesforce record Id → <b>must use a separate Graph connection</b> from the live Salesforce connector |
| Runtime       | Standalone .NET 10 Windows service · 981 automated tests                                                        |

## Signature capabilities

- **Filter-first:** partition pruning (skip dt=/region= folders) + 14 record-predicate operators + per-object row cap
- **Fail-closed scale guard:** an object with no effective filter refuses to crawl (would scan 150M rows) unless explicitly allowed
- 24-hour dt-watermark drives incrementals after the nightly sync lands
- Delegation token never logged; WebHDFS retry ladder; owner/group/public ACL
- **validate-config --strict is the gate:** all four config files (schema, filters, graph-schema, classification) validated up front; a produced-but-undeclared Graph property is a preflight ERROR, and the verdict does not depend on the CLASSIFICATION flag for any file that exists

## Before setup — key values to request

Request these (see the full guide for the complete checklist)

| Value / setting                                                                  | Ask from                        |
|----------------------------------------------------------------------------------|---------------------------------|
| <b>WebHDFS endpoint</b> (incl. /webhdfs/v1) or the mount path                    | Hadoop / BDH platform team      |
| A <b>read-only HDFS principal</b> (and private CA bundle if TLS-inspected)       | Hadoop / BDH platform team      |
| The Hive <b>partition layout</b> and object-to-path mapping                      | Hadoop / BDH platform team      |
| A <b>separate Graph connection</b> id + sign-off on the per-object filter policy | Connector operator + data owner |

## Key signals to monitor

| Metric / signal                         | Healthy looks like                                            |
|-----------------------------------------|---------------------------------------------------------------|
| Rows scanned vs matched vs pruned ratio | Most partitions pruned; scan is a small fraction of 150M      |
| Guard-refusal count                     | Zero in steady state; a spike means an object lost its filter |
| Watermark age                           | Under ~26h; older means the nightly sync is late              |
| ..._dead_letter_depth                   | Near zero                                                     |

## Top troubleshooting

| Symptom                                    | First action                                                                      |
|--------------------------------------------|-----------------------------------------------------------------------------------|
| An object is refused ("full-scan refused") | Add a real filter in filters.json; use ALLOW_FULL_SCAN only if genuinely intended |
| Partial crawl / deletion sweep suppressed  | An oversized file was skipped — handle that file; the sweep self-protects         |

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|                               |                                                                                         |
|-------------------------------|-----------------------------------------------------------------------------------------|
| Identity directory incomplete | Loud by design — the sync stops rather than apply coarse ACLs; fix the directory export |
| Watermark stale > 26h         | Check the nightly BDH→Hadoop sync job upstream                                          |

Full detail: **05\_Hadoop\_BDH\_Connector\_Guide.docx** and the Tenant & Common Concepts Guide (00). Metric prefix shown is per-connector.